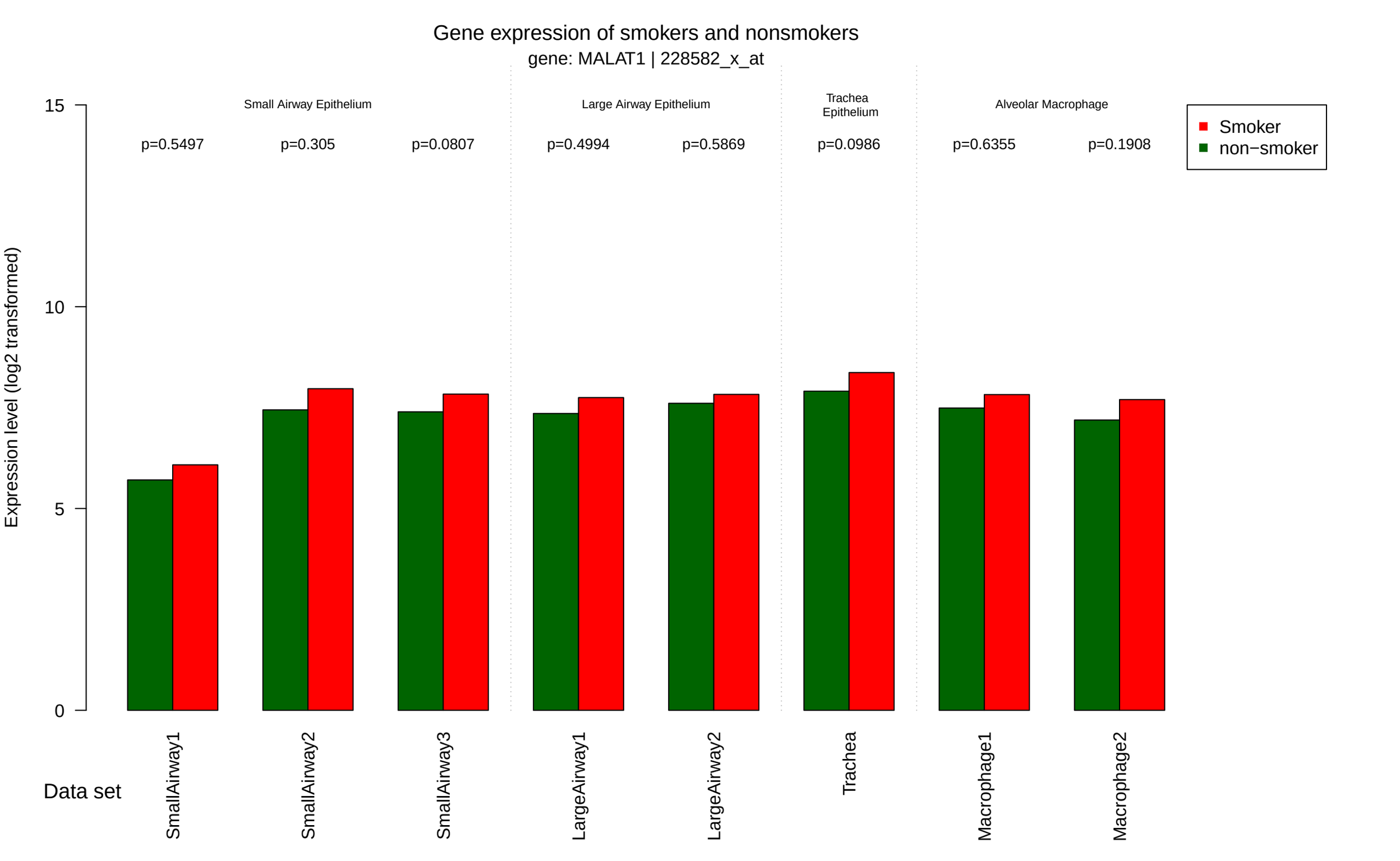
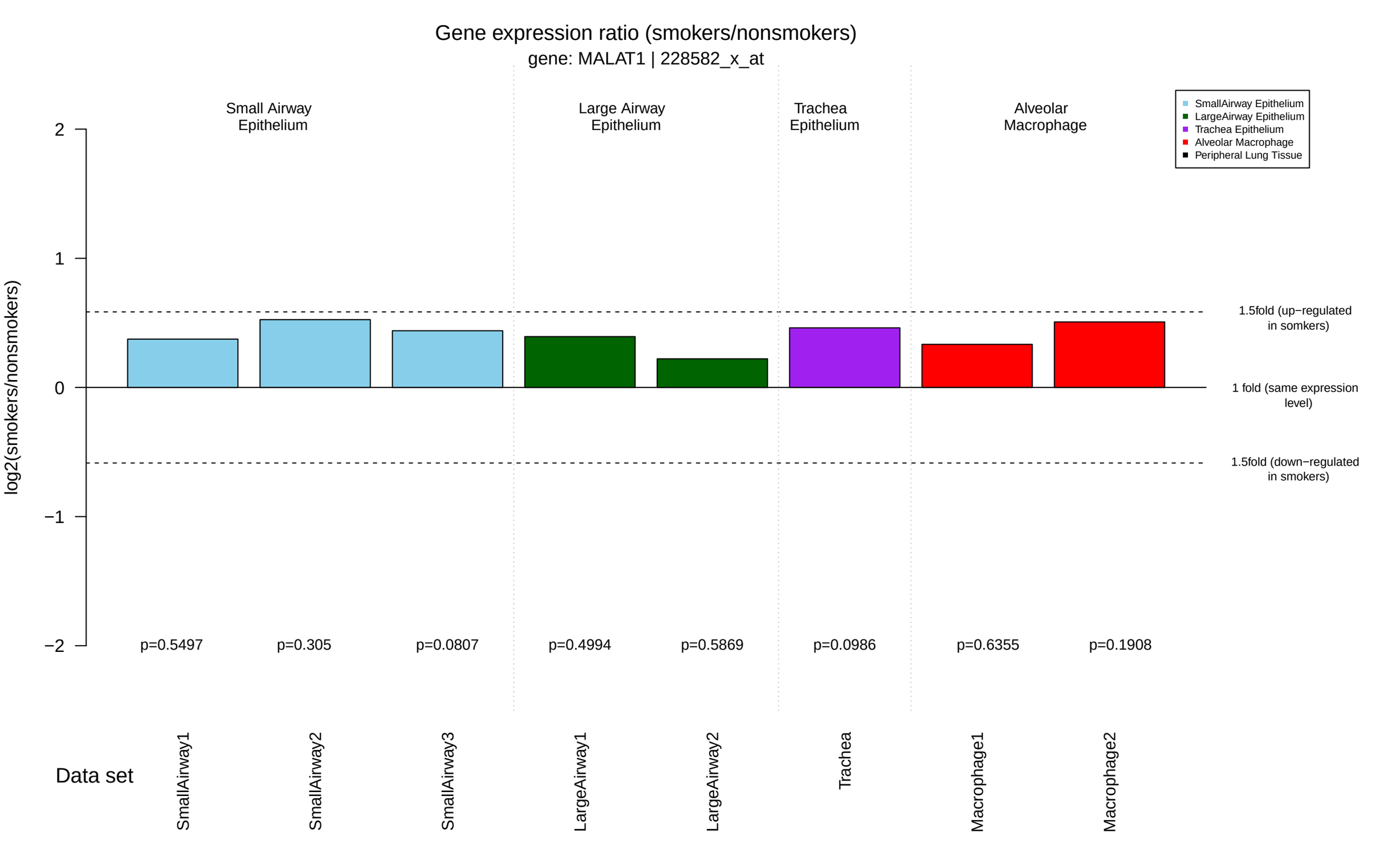
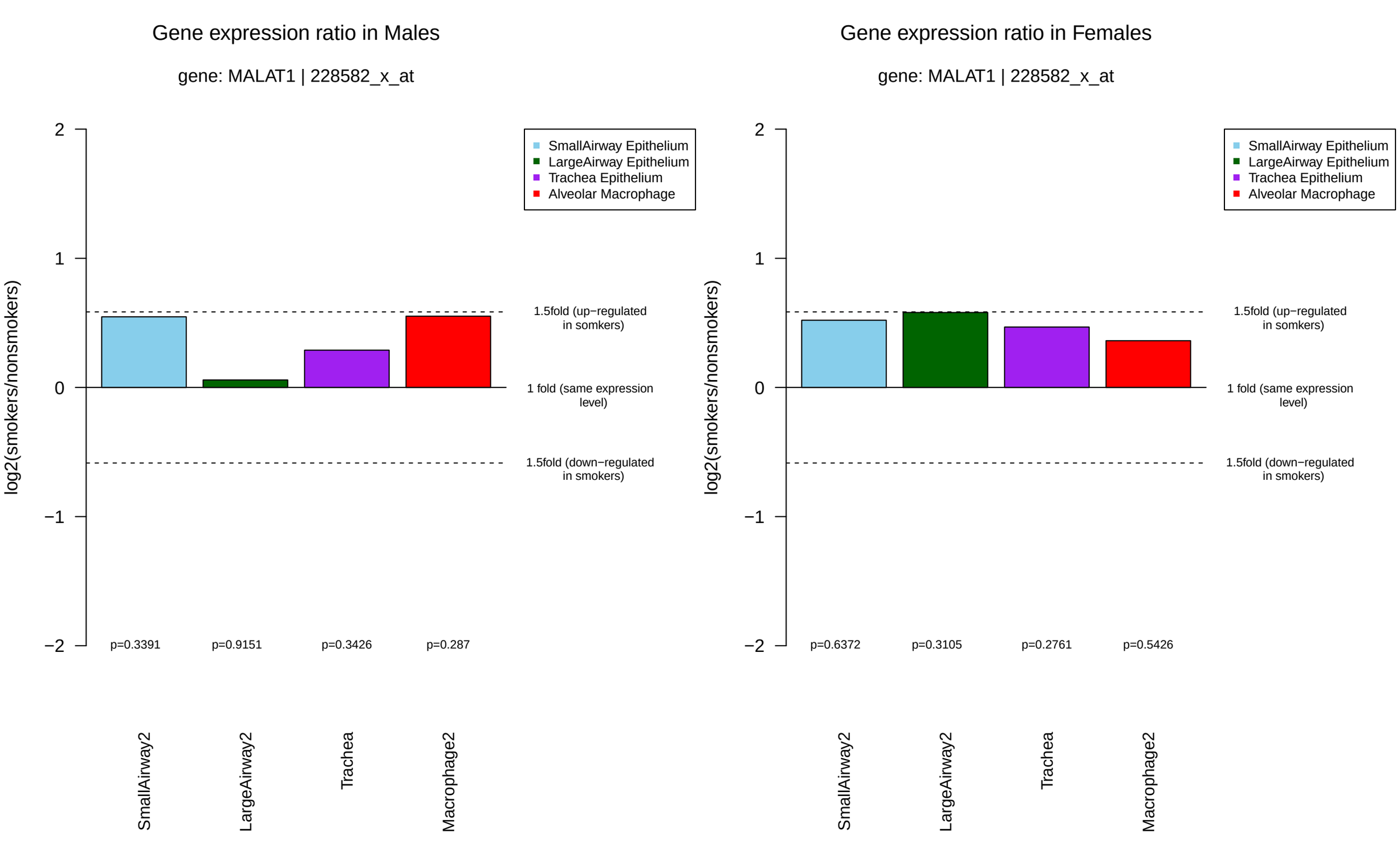
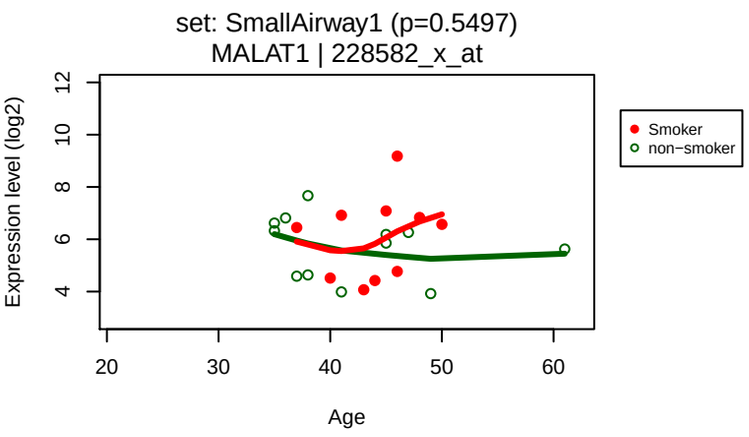
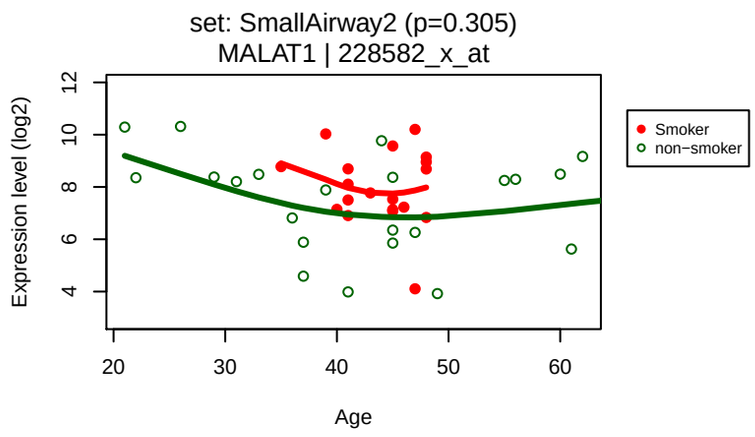
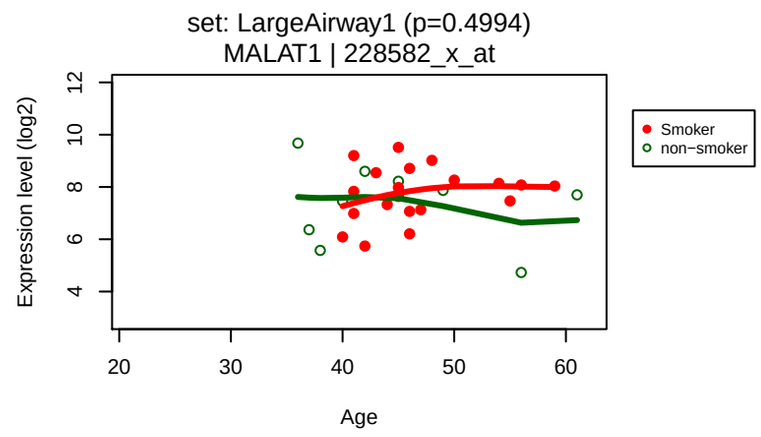

Gene expression of smokers and nonsmokers
gene: MALAT1 | 228582_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5497
p=0.305
p=0.0807
p=0.4994
p=0.5869
p=0.0986
p=0.6355
p=0.1908
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

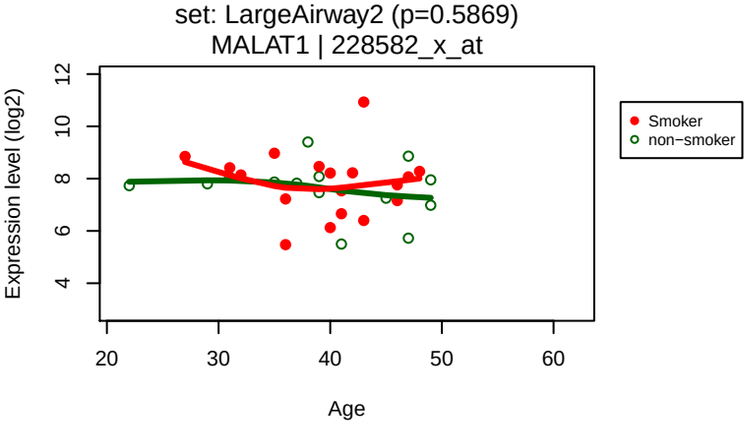

Gene expression ratio (smokers/nonsmokers)
gene: MALAT1 | 228582_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5497
p=0.305
p=0.0807
p=0.4994
p=0.5869
p=0.0986
p=0.6355
p=0.1908
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

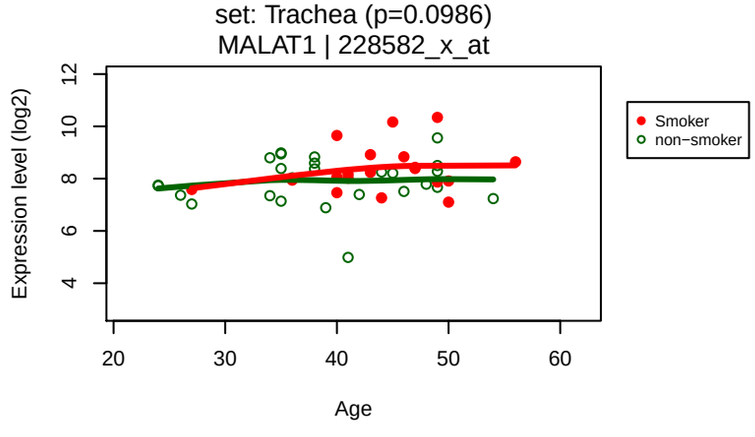

Gene expression ratio in Males
gene: MALAT1 | 228582_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3391
p=0.9151
p=0.3426
p=0.287
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MALAT1 | 228582_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6372
p=0.3105
p=0.2761
p=0.5426
SmallAirway2
LargeAirway2
Trachea
Macrophage2

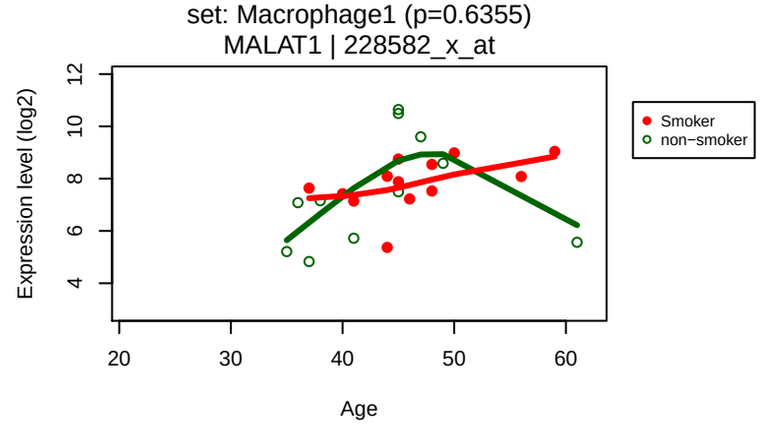

set: SmallAirway1 (p=0.5497)
MALAT1 | 228582_x_at
Expression level (log2)
Age
Smoker
non−smoker

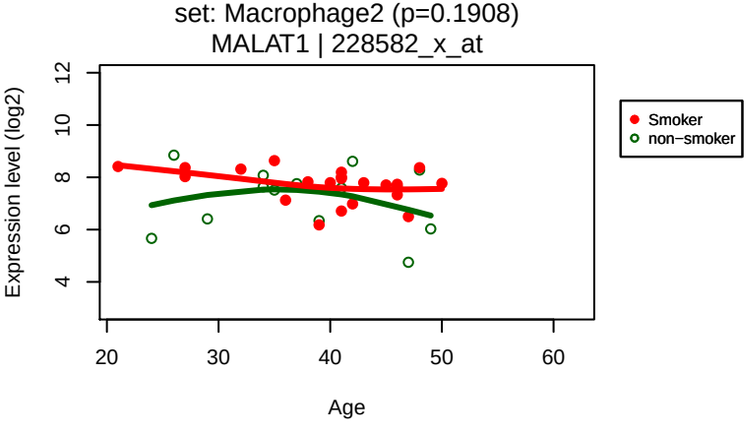

set: SmallAirway2 (p=0.305)
MALAT1 | 228582_x_at
Expression level (log2)
Age
Smoker
non−smoker

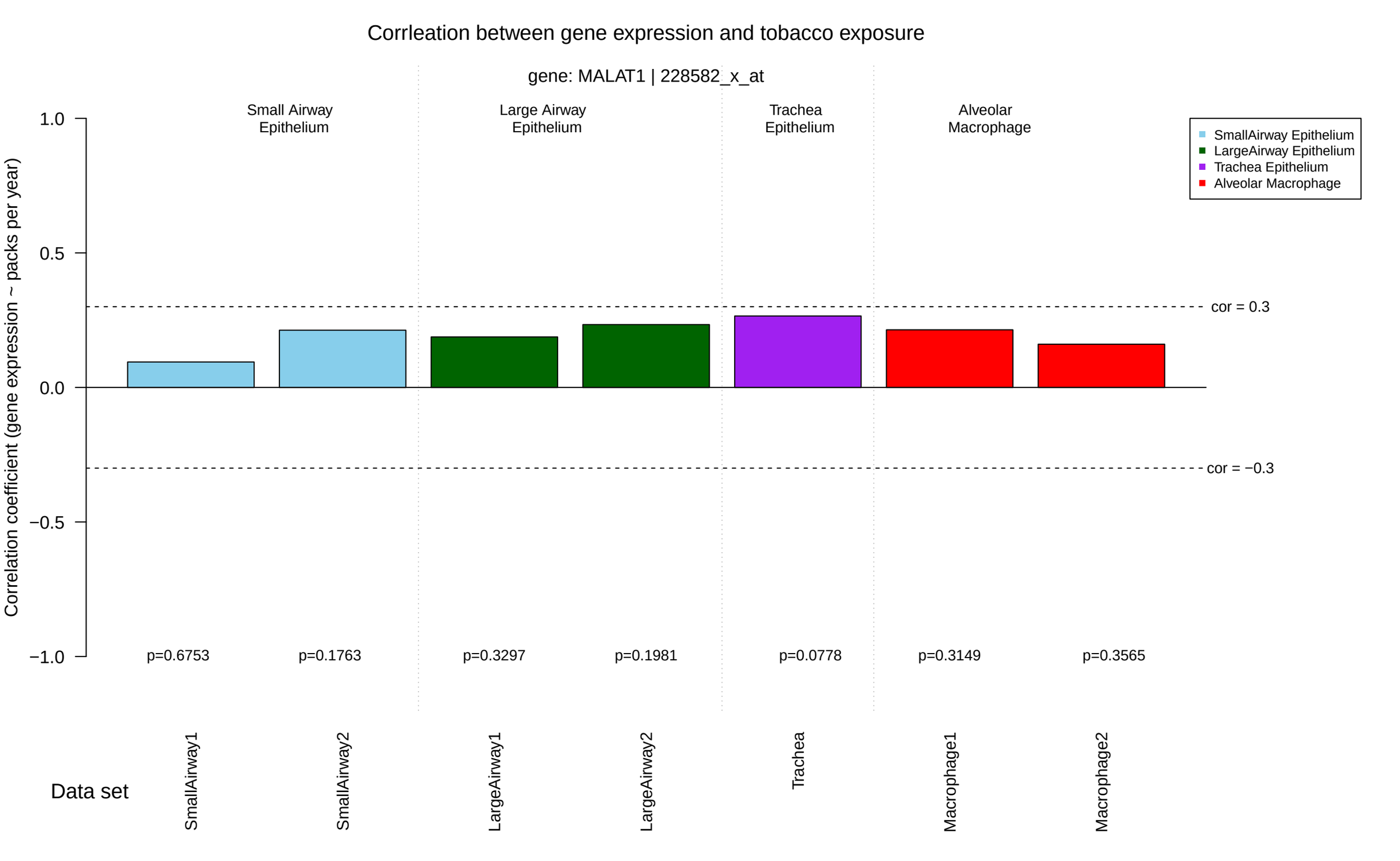

set: LargeAirway1 (p=0.4994)
MALAT1 | 228582_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5869)
MALAT1 | 228582_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0986)
MALAT1 | 228582_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6355)
MALAT1 | 228582_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1908)
MALAT1 | 228582_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MALAT1 | 228582_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6753
p=0.1763
p=0.3297
p=0.1981
p=0.0778
p=0.3149
p=0.3565
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2